JUN XIANG

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EDUCATION

Ph.D. in Mechanical Engineering Expected 2026
UC San Diego / San Diego State University
Advisor: Professor Jun Chen, Professor Sonia Martínez GPA: 3.5
Master of Automotive Engineering, Clemson University 2019 - 2021
Area of Study: Autonomous Vehicle GPA:3.7
Bachelor of Mechanical Engineering, North Carolina State University 2015 - 2019

EXPERIENCE

Research Assistant August 2021 - Present
UC San Diego/San Diego State University *San Diego, CA*

- Developed **reinforcement-learning and neural-network controllers** for high-dimensional nonlinear systems; formulated **Lyapunov and control-barrier conditions** and built an accelerated **auto_LiRPA/hybrid MILP** pipeline to verify closed-loop stability and safety.
- Developed **Barrier-Lyapunov-Agent**, an **AI-agent framework** for automatically training and verifying safe and stable neural-network controllers.
- Developed multiple machine-learning-based trajectory prediction frameworks using **LSTMs, GANs, Transformers, diffusion models, and vision-language models (VLMs)** for long-horizon forecasting.
- Designed **self-supervised and imitation-learning methods** to construct convex approximations of probabilistic reachable and obstacle sets from limited samples, supporting safety analysis under uncertainty.
- Developed **learning-accelerated A* and RRT* algorithms** for risk-aware motion planning; collected motion-capture data and conducted real-world flight experiments to validate autonomous-system behavior.

Research Assistant August 2019 - August 2021
Clemson University International Center for Automotive Research *Greenville, SC*

- Proposed a Q-Learning controller to find a control policy to control the vehicle's acceleration during the car following to maximize ride comfort. A paper has been published on SAE WCX.
- Applied GANs to predict pedestrian motion with the dataset from the Lyft Motion Prediction for Autonomous Vehicles Challenge and Waymo Motion Prediction Challenge.
- Applied CNNs to determine passenger's emotion and recognize passenger's command by passenger's voice.
- Collected data such as fuel efficiency, speed, and brake history from real autonomous vehicles with CAN.

Image Recognition Engineer Intern June 2019 - August 2019
Toshiba-APG Intelligent Drive Lab *China*

- Prepare a dataset of pictures of the rear view of the vehicles for deep learning.
- Performed testing and calibration for vehicle electronics such as sensors, radars, and cameras.

Mechanical Engineer Intern July 2017 - September 2017
Yuco Optics Corporation *Bohemia, NY*

SKILLS

Programming	Python, C++, MATLAB
Frameworks & Tools	PyTorch, TensorFlow, ROS, Git, Simulink
Planning & Control	A*, RRT*, MPC, LQR, Lyapunov stability, CBFs

PUBLICATIONS

Journals

- Xiang, J., Martinez, S, Chen, J. (2025). **Multi-Phase Reinforcement Learning for Formally Certified Safe and Stable Control.** *Automatica* (Submitted)
- Xiang, J., Chen, J. (2026). **Convex Approximation of Probabilistic Reachable Sets from Small Samples Using Self-supervised Neural Networks**. *IEEE Transactions on Systems, Man, and Cybernetics: Systems.*
- Xiang, J., Chen, J. (2025). **Data-driven probabilistic trajectory learning with high temporal resolution in terminal airspace**. *Journal of Aerospace Information Systems*
- Xiang, J., Liu, Y.C., Chen, J. (2022). **Hybrid Strategy with Multi-scale A* for Dynamic Planning of Multi-agent Drone Traffic**. *Journal of Guidance, Control, and Dynamics*
- He, Q., Xiang, J, et al. (2025) **3D-Printed programmable robotic skin with tunable toughness and surface-sensitive piezoelectric sensing**. *in process*

Conference Proceedings

- Xiang, J., Chen, J. (2026). **Barrier-Lyapunov-Agent: Training and Verifying Safe and Stable Neural Network Controllers** . *ICLR 2026* (submitted)
- Xiang, J., Chen, X.W, Chen, J. (2026). **Intention-Guided Airport Surface Trajectory Forecasting Using a Vision Language Model** . *AIAA Scitech 2026 Forum* (best student paper finalist)
- Xiang, J., Chen, J. (2026). **Context-Conditioned Diffusion Model for Long-Term Aircraft Trajectory Prediction Denoising from Pre-Generated Intention and Waypoints** . *IEEE International Conference on Systems, Man, and Cybernetics (SMC) 2026* (accpeted)
- Xiang, J., Chen, J. (2025). **Learning-accelerated RRT* Search for Risk-aware Path Planning**. *AIAA Aviation 2025*
- Xiang, J., Chen, J. (2024). **Transformer-based Heuristic for Advanced Air Mobility Planning**. *IEEE/AIAA Digital Avionics Systems Conference (DASC) 2024*
- Xiang, J., Chen, J. (2024). **Imitation Learning-based Convex Approximations of Probabilistic Reachable Sets**. *AIAA Aviation 2024 Forum*
- Xiang, J., Chen, J. (2024). **Learning-accelerated A* Search for Risk-aware Path Planning**. *AIAA Scitech 2024 Forum*
- Xiang, J., Essick, D., Bautista, L.G., Xie, J.F., Chen, J. (2023). **Landing Trajectory Prediction for UAS Based on Generative Adversarial Network**. *AIAA Scitech 2023 Forum*
- Xiang, J., Amaya, V., Chen, J. (2022). **Dynamic Unmanned Aircraft System Traffic Volume Reservation Based on Multi-Scale A* Algorithm**. *AIAA Scitech 2022 Forum*
- Xiang, J., Guo, L. (2022). **Comfort Improvement for Autonomous Vehicles Using Reinforcement Learning with In-Situ Human Feedback**. *SAE Technical Paper*
- Chen, X.W, Xiang, J., Chen, J. (2026). **Physically Consistent UAV Trajectory Prediction via Structured State-Space Representations and Autoregressive Transformers**. *AIAA Aviation 2026*
- Xu, Z., **Xiang, J.**., Chen, J., Dai, R. (2026). **Finite-Time Prediction Error Analysis for Neural Bilinear Koopman Models under Noisy Measurements**. *IEEE Conference on Decision and Control*
- Amaya, V, **Xiang, J.**, J, Chen, J. (2025), **Aviation Robustness Analysis for Learning Based Model for E-VTOL**. *AIAA Aviation 2026*
- Gault, J., **Xiang, J.**, Chen, J. (2023). **Safe Path Planning of UAV Based on Reinforcement Learning in Probabilistic Environments**. *AIAA Scitech 2023 Forum*